MGT 575 Generative AI for Business Final Project

YoSOM

A two-sided AI system for Yale SOM Dining and Ops

Aligning supply with student demand at Charley's Place and McNay Cafe

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Live: yosom.vercel.app

The problem

SOM dining runs on guesswork. Students feel unheard. Food gets thrown out at 2 PM.

~700

students eat on campus most days

Guesswork

and gut feel drive menu planning today

1000+

of meals served per week across two venues

Two specific pain points

Food waste at 2 PM

A large share of food prepared at peak hours gets tossed by closing time. Anyone who has walked past Charley's at 2 PM has seen it.

Performative surveys

Surveys happen rarely and feel performative. Students fill them out, or don't, and never see a result. Trust quietly erodes.

What we built

A two sided generative AI system. Students talk. Managers act.

STUDENT SIDE

Mobile chatbot

QR code or link • under a minute

- Pick venue: Charley's Place or McNay Cafe
- Tap preference: flavors, proteins, cuisines, dietary
- Vote on four option menu items
- Optional free text: what are you craving?
- See live feed of what others are voting for

MANAGER SIDE

Vendor dashboard

Desktop • five minutes a day

- Overview KPIs: sessions, satisfaction pulse, forecast
- Demand forecast with event and exchange multipliers
- AI Ops Brief: add, keep, drop with rationale
- Raw Data audit surface (every claim is checkable)
- Poster generation for promotional visuals

Try it live

Scan to open the student app on your phone.

SCAN TO TRY IT NOW

yosom.vercel.app

Mobile first • no app install required

STUDENT APP

yosom.vercel.app

What you'll scan to

MANAGER DASHBOARD

yosom.vercel.app/manager

Desktop, five sub pages

SOURCE CODE

github.com/yuanlee218/YoSOM

README + SOP, ready to clone

Architecture

Single Next.js app. Two interfaces, one shared database (MongoDB)

Student chat → **MongoDB** → **Aggregator** → **Claude / DALL E** → **Manager dashboard**

Parallel aggregation pipelines. Stateless AI calls. API keys stay server-side

Aggregate + Score

Six Mongo aggregations in parallel. Satisfaction pulse blends three signals into one 0 to 100 score with a trend label.

Forecast

Baseline servings × survey × event × exchange multiplier. Per day prediction with confidence band.

Synthesize + Reconcile

Claude returns structured JSON. Server stamps real cost numbers from the cost table. Hallucinations caught before display.

Three places we used generative AI

Pick the smaller, faster model for schema constrained tasks.

01

Claude Haiku 4.5

Student chat + tag extraction

- Natural classification > rigid rules
- Converts chat → structured preferences

02

Claude Haiku 4.5

Ops brief synthesis

- Switched from Sonnet → faster (3–5x)
- Same quality, no timeout

03

OpenAI DALL E 3

Promotional posters

- High-quality visuals in ~30s
- Optional: gpt-image-1 for text rendering

How the ops brief gets made

Six step pipeline: deterministic computation + one structured generation call.

1	2	3	4	5	6
Aggregate	**Score**	**Forecast**	**Synthesize**	**Reconcile**	**Cache**
Six Mongo pipelines in parallel One unified dataset	Satisfaction pulse 0–100 Survey multiplier in [0.85, 1.15]	Baseline × survey × event × exchange Per day prediction	Claude returns structured recommendations	Server stamps real costs from cost table Model outputs validated against source data	5-minute in-memory cache

If Claude fails (rate limit, network), the pipeline falls through to a deterministic fallback. The manager always sees a complete brief.

One honest limition: selection bias

Only students who scan a QR code and have 90 seconds participate. Their voice becomes "students."

WHO PARTICIPATES

132

≈ 1 in 5 SOM students

ALL OF THEM:

- Engaged enough with food to vote
- Comfortable typing in English
- Had 90 seconds to spare
- Trust QR codes from strangers

WHO STAYS SILENT — and shapes nothing

568

≈ 4 in 5 SOM students — and we don't know who

INCLUDING:

- Introverted students who don't speak up
- International students still hesitant in English
- Students who already gave up on Charley's
- Anyone too busy to scan or type

The silent 568 still eat what the loud 132 voted for. ***That's the bias.***

Possible Solution: *sample-size confidence label on the headline + **"who's missing" demographic flag** once we have any signal at all.*

Section 5

The bigger vision

YoSOM today is a dining tool.

YoSOM in eighteen months, if we earn the runway,
is the operating system of Evans Hall.

A multiagent vision for Evans Hall

Eight specialized agents. One shared event bus. Humans in the loop on every dollar.

LIVE

Dining

Shipped. Demand forecasting, menu briefs, posters.

Procurement

Drafts purchase orders. Flags duplicate spend across departments.

Energy

HVAC + lighting telemetry crossed with the room calendar.

Events

Master calendar. Resolves room conflicts. Drafts run of show docs.

Facilities

Triages work orders. Groups nearby tickets. Briefs technicians.

Career Office

Recruiter visits end to end: rooms, catering, prep notes.

Wayfinding

Lobby chatbot. Logs missing signage as a Facilities feed.

ROUTER

Orchestrator

Routes the event bus. One executive brief in the morning.

Shared event bus. Each agent publishes typed events. The Orchestrator reasons over them uniformly.

The shared memory layer

Karpathy's LLM Wiki pattern as the institutional memory of Evans Hall.

THE GAP

Each agent is stateful inside its domain. The network has no shared memory of the building.

When the Career Office director who knows Goldman wants seafood leaves SOM, that knowledge leaves with her. When the BMS technician who figured out which rooms have phantom heating retires, the next person rediscovers it from scratch.

> *"Obsidian is the IDE; the LLM is the programmer; the wiki is the codebase."*
>
> — Andrej Karpathy, April 2026

FOUR WIKIS EMERGE NATURALLY

Vendor wiki

Procurement. One page per supplier with contracts, history, owner, contradictions over time.

Space wiki

Facilities + Energy. One page per room: capacity, AV, recurring bookings, energy quirks, work orders.

Recruiter wiki

Career Office. One page per firm: preferences, partners, post visit themes, candidate outcomes.

Student voice wiki

Dining. Each menu item, dietary cluster, event week becomes a page that compounds across semesters.

Rollout and the ask

One year sequence. Highest ROI agents first.

PHASED ROLLOUT

Q1

Harden YoSOM Dining

Real cost data, real volume, closed loop measurement. Stand up event bus.

Q2

Energy Agent

Highest ROI, cleanest data. BMS feed plus room calendar ships v1.

Q3

Events + Procurement

The two agents that most amplify Dining and Energy value.

Q4

Facilities, Career, Wayfinding

Student facing agents. Higher trust bar. Ship after orchestration is proven.

THREE RISKS WE'RE WATCHING

- Governance: agents that spend money need approval ladders, audit trails, kill switches
- Integration: BMS, procurement portals, room calendars were not built for API access
- Organizational: value unlocked only if SOM ops lets agents draft work humans used to do

THE ASK

- A real pilot conversation with SOM Dining
- One quarter of historical demand and cost data
- Permission to talk to the BMS team about a read only feed

If today's demo is interesting, we want a real conversation.